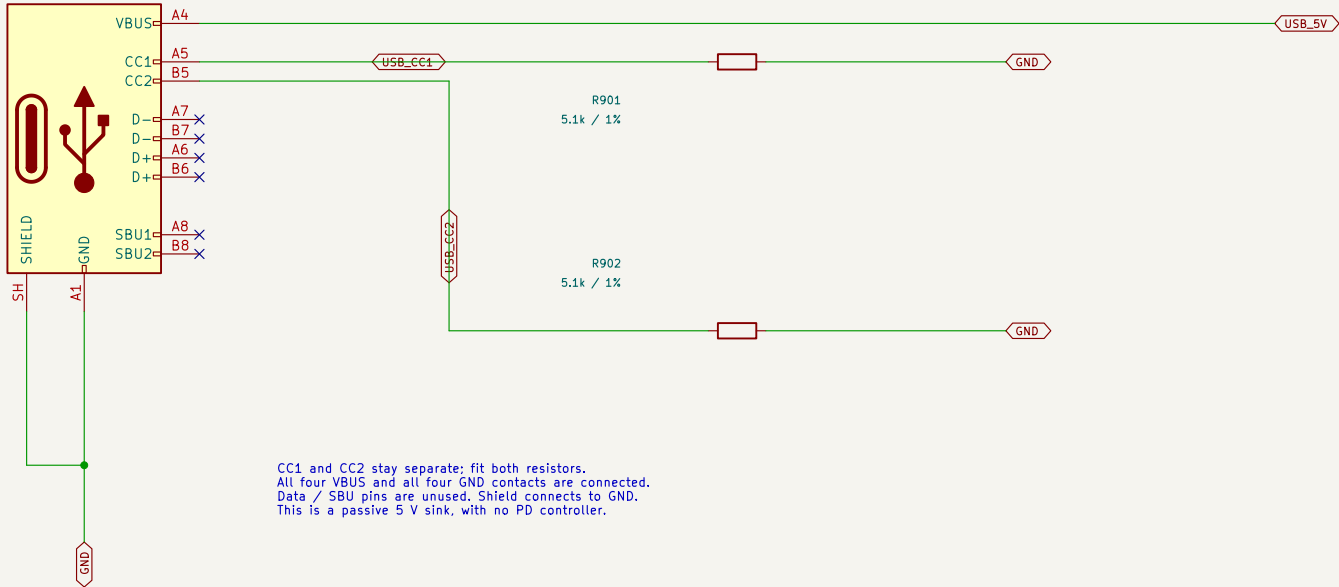


USB INPUT | GCT USB4720-03-A DAUGHTERBOARD

5 V charging from USB-A or USB-C. Separate 0.60 mm USB PCB; main charger PCB remains 1.60 mm.

01 GCT SOCKET + TWO CC RESISTORS

J901
GCT USB4720-03-A



CC1 and CC2 stay separate; fit both resistors.
All four VBUS and all four GND contacts are connected.
Data / SBU pins are unused. Shield connects to GND.
This is a passive 5 V sink, with no PD controller.

02 TWO-WIRE OUTPUT TO CHARGER

J902
TO MAIN PCB J101



J902 pin 1 -> positive wire -> J101 pin 1
J902 pin 2 -> return wire -> J101 pin 2

J901 / R901 / R902 / J902 belong to the USB PCB.
The main charger retains its two-wire input.
Use soldered leads with enclosure strain relief;
wire size and final mounting remain to be checked.

03 CABLES, ORIENTATION & CURRENT

USB-C charger + C-to-C cable: separate Rd on CC1 / CC2
allows source attachment in either plug orientation.
Test both ends flipped, including an electronically marked cable.

USB-A charging adapter + A-to-C cable uses the same input.
The compliant legacy cable supplies Rp; do not add Rp here.

BQ25895 D+/D- remain unused; nominal 500 mA INPUT setting.
Battery charge current depends on available power and system load.
No 9/12/20 V PD request or 1.5/3 A detection is implemented.

04 MECHANICS & VERIFICATION

GCT drawing: 0.60 +/- 0.10 mm PCB and edge cutout.
Do not fit USB4720 directly to the 1.60 mm main board.
Provide rigid support against cable insertion forces.
IP67 connector does not certify the enclosure; charge dry.

Before power: verify CC1-GND and CC2-GND -5.1k.
With ARM OPEN: verify 5 V in all cable orientations,
including no battery / host off. Then qualify loaded
voltage, input current and inrush. Battery charging
tests follow cell / holder / thermistor qualification.

Engineering review draft – not released for manufacture

Trimix

Sheet: /

File: Trimix_USB_Input.kicad_sch

Title: Trimix USB daughterboard – GCT USB4720-03-A

Size: A3

Date: 2026-09-06

Rev: A2.2

KiCad E.D.A. 10.0.6

Id: 1/1